

Project Status Report

This document outlines the current state of the **CS Compliance Dashboard** project, highlighting completed features, pending components, and unresolved bugs.

1. Completed Work

Backend & Core

- **Database Migration:** Migrated database structure from PostgreSQL to **MongoDB** using **Beanie ODM** (Object Document Mapper) and Motor async driver.
- **Unified Regulatory Library:** Integrated a searchable compliance search engine inside *backend/app/services/regulatory_library.py* that merges scraped JSON updates from multiple regulators.
- **Rule Engine & Scheduler:** Added background task scheduling using **APScheduler** to automatically run compliance checks against assigned client companies.
- **API Endpoints:**
 - */auth*: Sign-in, sign-up, and JWT-token generation.
 - */clients*: Create, read, update, delete, assigned tasks, and audit logs.
 - */tasks*: Search, assign, complete, and reopen tasks.
 - */regulatory-updates*: Query and filter scraped updates.
 - */admin*: User directory and system statistics.

Web Scraping Services (*backend/app/services/*)

- **MCA Scraper:** Crawls the Ministry of Corporate Affairs acts, rules, and e-books.
- **RBI Scraper:** Logs into the RBI portal, strips webdriver signatures, and extracts Master Directions, Circulars, Notifications, Press Releases, and FAQs.
- **Labour Laws Scraper:** Crawls the Ministry of Labour & Employment consolidated directory using Playwright in headed mode to collect Labour Codes, Rules, and Notifications.
- **ICSI Scraper:** Scrapes SS-1 through SS-10, Guidance Notes, FAQs, and Amendments.
- **Unified Data Outputs:** Generated and placed standard compliance datasets in *backend/* for search indexing: *mca_scraped_data.json*, *rbi_scraped_data.json*, *labour_scraped_data.json*, *icsi_scraped_data.json*, and *atlas_vayana_scraped_data.json*.

Frontend UI (*frontend/*)

- Built a responsive dashboard with Tailwind CSS and React.
- **Modules Completed:** Client Firm Management, Tasks Board, Regulatory updates search library, and Admin & Partner metrics panel.

2. Pending Work

- **IP India Scraper Integration:** Teammates removed *ipindia_scraped_data.json* from git tracking, and it is currently excluded from the search library. If Intellectual Property (patents, trademarks, and copyright) tracking is still needed for clients, this must be re-added to *DATA_FILES* in *regulatory_library.py*.
- **Secretarial Standards SS-5 to SS-10:** Finalized official texts/PDFs for SS-5 (Minutes) to SS-10 (Board's Report) are currently integrated as draft template definitions. When these are formally updated on the ICSI portal, live PDF parsing should be added.

3. Remaining Bugs & Issues to Solve

- **Local MongoDB Startup Dependency:** Running the API or database seed script (*seed.py*) locally on a machine without a running MongoDB service on port 27017 causes a *ServerSelectionTimeoutError* crash. Keep the MongoDB server started locally, or utilize Docker Compose to launch the database container automatically.
- **Unused Import in App.jsx:** *TaskDetail* is imported as a page route in *frontend/src/App.jsx* but not registered in the *Routes* hierarchy (since task details are rendered as interactive drawers directly inside *TaskList.jsx* and *Dashboard.jsx*). The unused import should be cleaned up.

4. AI Integration & Training Strategy

Ingestion & Training Pipeline

- **Dataset Utilization:** The scraped JSON datasets (MCA, RBI, Labour, and ICSI) will serve as the core knowledge base.
- **RAG (Retrieval-Augmented Generation) Architecture:**
 - *Text Chunking:* Chunk the full text of rules into semantic passages (e.g. 500-character chunks with overlap).
 - *Vector Embeddings:* Generate vector representations using models like *text-embedding-004*.
 - *Vector Search Store:* Save the embeddings inside a MongoDB vector search index or a dedicated vector database.
 - *Contextual Retrieval:* On user query, retrieve relevant passages and provide them to an LLM (such as Gemini 1.5 Pro) to compile grounded, audit-ready compliance answers.

Expected Capabilities from the Integrated AI

- **Compliance QA & Research:** Answer complex queries on filing dates, exemptions, extensions, and penal consequences.
- **Draft Form Validation:** Match pre-filled draft forms (such as MGT-7 or AOC-4 documents) against current circulars to identify discrepant rules.
- **Intelligent Alerting:** Summarize newly scraped circulars and tag the specific client firms that are affected by the changes.